DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

Subject : Operating Systems

Class: B.E (CSE)
Semester: IV
Academic Year: 2026

Max Marks: 30
Duration:1 ½ hr
Date : -05-2026

		Marks	BT	CO
	PART A Answer All Questions (Q5 x 2M = 10 M)			
1.	What are the four conditions that a deadlock can arise?	3	3	1
2.	What is logic bomb? How it may manifest itself?	2	3	1
3.	Define Beladoy's anamoly.	2	2	1
4.	Explain the dynamic linking.	3	1	1
	PART B Answer Any Two Questions (02 x 10M = 20M)			
5.	Illustrate the use of the banker's algorithm, consider a system with five processes Po through P4 and three resource types A, B, and C. Resource type A has ten instances, resource type B has five instances, and resource type C has seven instances. *Allocation* *Max* *Available* A B C A B C A B C P_0 0 1 0 7 5 3 3 3 2 P_1 2 0 0 3 2 2 P_2 3 0 2 9 0 2 P_3 2 1 1 2 2 2 P_4 0 0 2 4 3 3 Check whether the system is in safe state or not . Justify a) Suppose now that process P1 requests one additional instance of resource type A and two instances of resource type C. Whether this request be immediately granted or not. b) a request for (3,3,0) by P4 can be granted or not. Justify	10	3	1
6.	a) Consider the below reference string, illustrate the Optimal and LRU page replacement algorithms. 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1	5	3	1
	b) Describe the process of encryption. What is public key encryption?	5	2	1
7.	a) Explain the implementation of access matrix.	5	3	1
	b) What are the steps required to handle a page fault in demand paging?	5	1	1